

May 7th, 2025

Dear Director of Research,

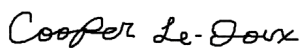
This report presents our proposed process for producing ϵ -caprolactam from biomass-derived phenol for use in nylon production. The process begins with phenol and utilizes a hydrogenation reaction to achieve a product purity of at least 98.5%. Included in this report are key aspects of our process, such as our process flow diagram, equipment lists, facility layout, cost summary, and any environmental, health and safety considerations. We kindly request a review of these materials in our report, and our recommendations for moving forward with this process. We welcome any feedback that could help us further refine and strengthen our project. We would also like to extend our gratitude to Dr. Brewer and Dr. Andersen at New Mexico State University for their guidance during the development of this work.

Signed,



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Lina Khiyara



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Cooper Ledoux



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Taylor Lee



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Estrella Sedillo



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Production of ϵ -Caprolactam Using Phenol Derived from Biomass

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Executive Summary

This report presents the design and economic evaluation of a chemical plant to produce ϵ -caprolactam from biomass-derived phenol, providing a renewable alternative to the current petroleum-based production route in nylon manufacturing. Phenol can be obtained by chemically converting lignin found in lignin cellulosic biomass, however this lignin conversion and pretreatment steps were not included within this project scope, and it was assumed that pure phenol was obtained as the starting material for this process. The comparison case for this report involved comparing nickel-based versus palladium-based catalysts for the hydrogenation of phenol reaction. Ruthenium-based catalysts were also initially considered but were quickly ruled out due to the price being significantly higher than nickel. The economic evaluation showed a net present value (NPV) of 18.4 million USD and -54.1 million USD for the palladium and nickel catalysts, respectively, confirming palladium is the preferred catalyst despite higher upfront costs. The process is expected to be in Albuquerque, NM 87120, run for 15 years, have a production rate of 544,000 US tons per year, and produce ammonium sulfate as a co-product which can be treated then sold to fertilizer companies. Modeling this process in Aspen confirmed that a pilot study will need to be done to determine optimal reactor pressures, namely for the plug flow reactor (PFR), as theoretical literature values do not produce successful operating conditions in the simulation. Based on economic performance and technical feasibility, management is recommended to proceed with the palladium-based process and begin process optimization and life cycle assessment in the next phase.

Process Flow Diagram and Process Description

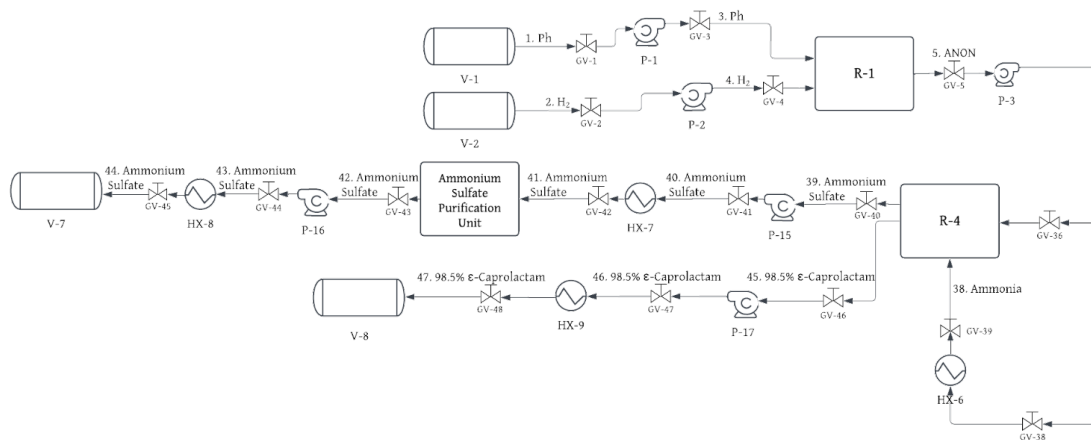


Figure 1A: PFD showing the hydrogenation reactor (R-1), ammonia reactor (R-4) used to treat the final product, and the ammonium sulfate purification unit; where Ph: phenol, H₂: hydrogen, and ANON: cyclohexanone

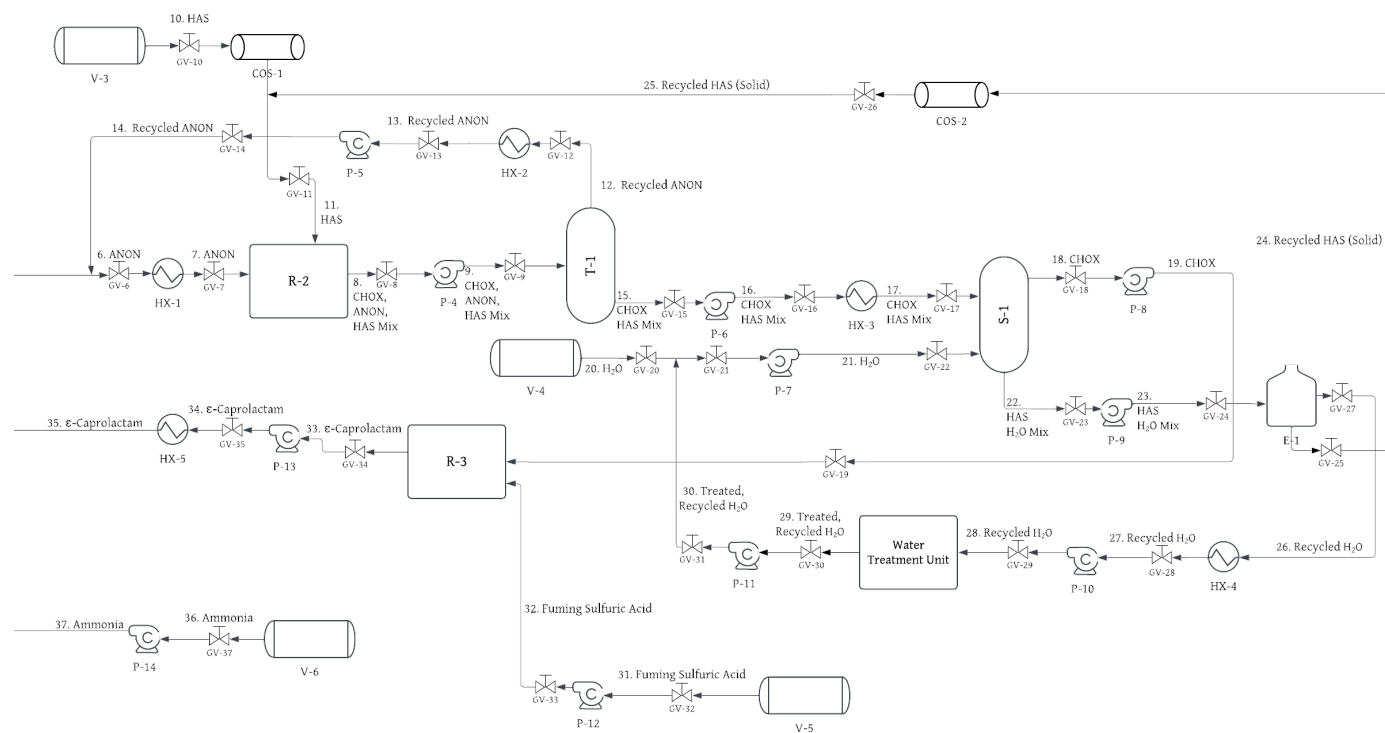


Figure 1B: PFD showing the CHOx formation reactor (R-2), CHOx separations units (T-1 and S-1), evaporator used to obtain recycled HAS (E-1), water treatment unit, and the Beckmann rearrangement reactor (R-3) that forms ϵ -caprolactam; where HAS: hydroxylamine sulfate, ANON: cyclohexanone, CHOx: cyclohexanone oxime, and H₂O: water

Table 1: Stream summary table that corresponds to PFD shown in Figure 1

Stream number	Chemical species	Temperature (C)	Pressure (bar)
1	Phenol	20	1
2	Hydrogen	20	1
3	Phenol	20	1
4	Hydrogen	20	1
5	ANON	450	2
6	ANON	450	2
7	ANON	85	2
8	ANON, CHO _X , HAS mix	85	2
9	ANON, CHO _X , HAS mix	85	2
10	HAS	25	1
11	HAS	25	1
12	ANON	160	1
13	ANON	450	1
14	ANON	450	1
15	CHO _X , HAS mix	160	1
16	CHO _X , HAS mix	160	1
17	CHO _X , HAS mix	95	1
18	CHO _X	95	1
19	CHO _X	95	1
20	Water	25	1
21	Water	25	1
22	HAS, water mix	95	1
23	HAS, water mix	95	1
24	HAS, water mix	100	1
25	HAS, water mix	100	1
26	Water	100	1.3
27	Water	25	1
28	Water	25	1

29	Water	25	1
30	Water	25	1
31	FSA	25	1
32	FSA	25	1
33	Caprolactam	90	1
34	Caprolactam	90	1
35	Caprolactam	100	1
36	Ammonia	-33	1
37	Ammonia	-33	1
38	Ammonia	100	1
39	AS	100	2
40	AS	100	2
41	AS	85	1
42	AS	60	1
43	AS	60	1
44	AS	25	1
45	Caprolactam	100	1
46	Caprolactam	100	1
47	Caprolactam	85	1

Process Description

The proposed process begins with phenol (derived from biomass) that undergoes catalytic hydrogenation in Reactor 1 (R-1) with operating conditions of 450°C and 1-2 bar to produce cyclohexanone [1]. It was determined that a plug flow reactor (PFR) was the most suitable option for this reaction, as it allows for high conversion, continuous flow (ideal for industrial processes), and catalytic packing [2] – palladium catalyst in this case.

The produced cyclohexanone is fed into Reactor 2 (R-2), a continuous stirred tank reactor (CSTR) with operating conditions of 85°C and 1-2 bar [1], along with hydroxylamine sulfate to produce cyclohexanone oxime. A CSTR was selected for this reaction as it allows sufficient

mixing, ensuring uniform concentration and temperature throughout the reactor which helps minimize potential side reactions occurring, such as the formation of cyclohexanol. The resulting effluent from this reactor is a mixture of the cyclohexanone oxime and unreacted hydroxylamine sulfate and cyclohexanone which is fed into a distillation column (T-1). This is done such that this unreacted cyclohexanone is removed from the mixture, based on its significantly higher volatility than the other two components, and recycled back into our process to minimize waste. The bottoms of T-1, now only a mixture of hydroxylamine sulfate and cyclohexanone oxime, is then fed into a centrifugal phase separator (S-1), along with water. This is done because hydroxylamine sulfate is readily soluble in water [3], while cyclohexanone oxime is not [4], allowing purified cyclohexanone oxime to be obtained which is needed for the remainder of the process. The hydroxylamine sulfate and water mixture is fed into an evaporator (E-1) so that the water and the hydroxylamine sulfate are separated. The water is condensed then treated and recycled back into the process. The hydroxylamine sulfate is also recycled back into the process to minimize waste and reduce raw material costs.

The purified cyclohexanone oxime is fed into Reactor 3 (R-3), a CSTR with operating conditions of 90°C and 1-2 bar [1], along with fuming sulfuric acid to produce ϵ -caprolactam via Beckmann rearrangement [5]. A CSTR was chosen as it allows sufficient mixing, ensuring complete mixing of the cyclohexanone oxime with the fuming sulfuric acid, improving reaction rate and yield, and because it supports continuous flow which is ideal for continuous industrial processes. The produced ϵ -caprolactam from this reaction then needs to be purified to meet industry and customer standards.

The crude ϵ -caprolactam that is obtained from the Beckmann rearrangement reaction will likely contain impurities such as unreacted cyclohexanone oxime, residual sulfuric acid, and water and trace minerals - all of which are impurities that need to be removed. Therefore, the final step of this process is purification. This is done by feeding the effluent from R-3 into Reactor 4 (R-4), a CSTR with operating conditions of 100°C and 1-2 bar, along with ammonia. A CSTR was chosen as it ensures ammonia is evenly distributed throughout the ϵ -caprolactam solution, ultimately neutralizing impurities. The ammonia fed into R-4 treats the product such that a yield of up to 98.5% pure ϵ -caprolactam is obtained [6], along with ammonium sulfate as a co-product. This co-product is purified such that it meets the standards of the fertilizer industry

and can be sold. After evaluation, it was determined that this industrial process facility, if ran 365 days per year, will produce approximately 544,000 US tons of ε-caprolactam per year. All stream information correlating to Figure 2, including temperature, pressure, and mass flow rates of each component can be found in Table 2

Process Flow Diagram Generated in Aspen

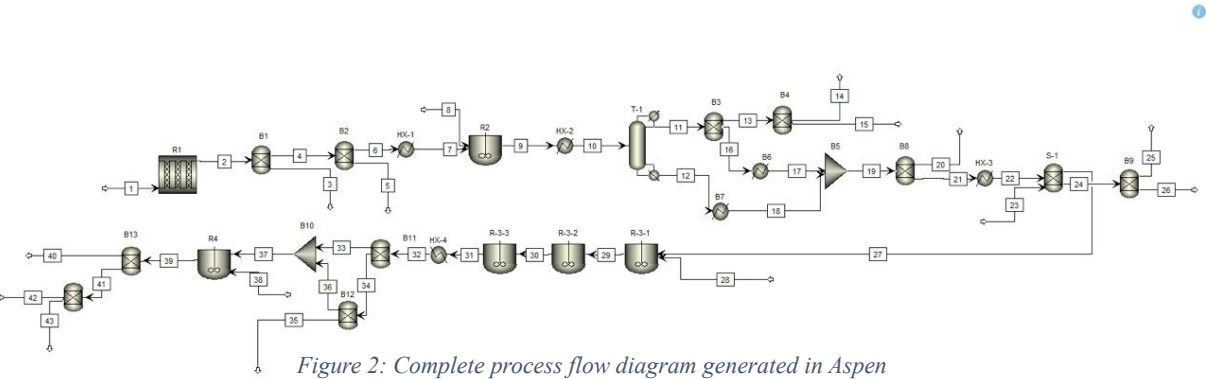


Figure 2: Complete process flow diagram generated in Aspen

Stream Summary Table

Table 2: Stream summary table corresponding to Aspen-generated PFD shown in Figure 2, including mass balance information (flowrates), temperatures, and pressures

Stream number	Chemical species	Temperature (C)	Pressure (bar)	Mass flow rate of chemical species (kg/hr)	Total mass flow rate (kg/hr)
1	Phenol, Hydrogen mix	450	35	Phenol: 60763.0 Hydrogen: 2651.0	63413.6
2	Phenol, Hydrogen, ANON mix	924	35	Phenol: 19.4 Hydrogen: 48.6 ANON: 63345.6	63413.6
3	Phenol	924	35	19.4	19.4

4	Hydrogen, ANON mix	924	35	Hydrogen: 48.6 ANON: 63345.6	63394.1
5	Hydrogen	924	35	48.6	48.6
6	ANON	924	35	63345.6	64456.6
7	ANON	85	35	63345.6	63345.6
8	HAS	85	35	51703.2	51703.2
9	ANON, HAS, Water, CHOX mix	85	35	ANON: 0.0 HAS: 30384.7 Water: 11627.6 CHOX: 73036.5	115049.0
10	ANON, HAS, Water, CHOX mix	200	35	ANON: 0.0 HAS: 30384.7 Water: 11627.6 CHOX: 73036.5	115049.0
11	ANON, HAS, Water, CHOX mix	227.8	35	ANON: 6.1e-05 HAS: 15552.7 Water: 5456.9 CHOX: 0.1	21009.0
12	ANON, HAS, Water, CHOX mix	259.9	35	ANON: 0.0 HAS: 14832.0 Water: 6170.7 CHOX: 73036.4	94039.0
13	ANON, Water, CHOX mix	227.8	35	ANON: 6.1e-05 Water: 5456.9 CHOX: 0.1	5457.0
14	Water	227.8	35	5456.9	5456.9
15	ANON, CHOX mix	227.8	35	ANON: 6.1e-05 CHOX: 0.1	0.1
16	HAS	227.8	35	15552.7	15552.7
17	HAS	95	35	15552.7	15552.7

18	ANON, HAS, Water, CHOX mix	95	35	ANON: 0.0 HAS: 14832.0 Water: 6170.7 CHOX: 73036.4	94039.0
19	ANON, HAS, Water, CHOX mix	87.1	35	ANON: 0.0 HAS: 30384.7 Water: 6170.7 CHOX: 73036.4	109592.0
20	ANON	87.1	35	0.0	0.0
21	HAS, Water, CHOX mix	87.1	35	HAS: 30384.7 Water: 6170.7 CHOX: 73036.4	109592.0
22	HAS, Water, CHOX mix	95	35	HAS: 30384.7 Water: 6170.7 CHOX: 73036.4	109592.0
23	Water	25	35	57.6	57.6
24	HAS, Water mix	94.8	35	HAS: 30384.7 Water: 6228.3	36613.0
25	Water	94.8	35	6228.3	6228.3
26	HAS	94.8	35	30384.7	30384.7
27	CHOX	94.8	35	73036.4	73036.4
28	Water, SO ₃ , FSA mix	25	35	Water: 12100.0 SO ₃ : 13902.3 FSA: 41706.9	67709.2
29	Water, CHOX, FSA, SO ₃ mix	90	35	Water: 8972.0 CHOX: 73036.4 FSA: 58737.4 SO ₃ : 1.6e-09	140746.0
30	Water, CHOX, H ₃ O ⁺ , FSA,	90	35	Water: 0.0 CHOX: 73036.4	140746.0

	SO ₃ , HSO ₄ - mix			H ₃ O ⁺ : 9473.5 FSA: 9892.5 SO ₃ : 1.6e-09 HSO ₄ ⁻ : 48343.1	
31	Water, CHO _X , H ₃ O ⁺ , H ⁺ , Caprolactam, FSA, SO ₃ , HSO ₄ ⁻ mix	90	35	Water: 8971.8 CHO _X : 16681.6 H ₃ O ⁺ : 0.0 H ⁺ : 501.7 Caprolactam: 56354.8 FSA: 9892.5 SO ₃ : 1.6e-09 HSO ₄ ⁻ : 48343.1	140746.0
32	Water, CHO _X , H ₃ O ⁺ , H ⁺ , Caprolactam, FSA, SO ₃ , HSO ₄ ⁻ mix	100	35	Water: 8971.8 CHO _X : 16681.6 H ₃ O ⁺ : 0.0 H ⁺ : 501.7 Caprolactam: 56354.8 FSA: 9892.5 SO ₃ : 1.6e-09 HSO ₄ ⁻ : 48343.1	140746.0
33	Caprolactam	100	35	56354.8	56354.8
34	Water, CHO _X , H ₃ O ⁺ , H ⁺ , FSA, SO ₃ , HSO ₄ ⁻ mix	100	35	Water: 8971.8 CHO _X : 16681.6 H ₃ O ⁺ : 0.0 H ⁺ : 501.7 FSA: 9892.5 SO ₃ : 1.6e-09 HSO ₄ ⁻ : 48343.1	84390.8

35	Water, CHO _X , H ₃ O ⁺ , H ⁺ , SO ₃ , HSO ₄ ⁻	100	34	Water: 8971.8 CHO _X : 16681.6 H ₃ O ⁺ : 0.0 H ⁺ : 501.7 SO ₃ : 1.6e-09 HSO ₄ ⁻ : 48343.1	74498.3
36	FSA	100	35	9892.5	9892.5
37	Caprolactam, FSA mix	92	35	Caprolactam: 56354.8 FSA: 9892.5	66247.4
38	Ammonia	100	35	5000.0	5000.0
39	Caprolactam, Ammonia, FSA, AS mix	100	35	Caprolactam: 56354.8 Ammonia: 1564.5 FSA: 7.4e-06 AS: 13328.0	71247.4
40	Caprolactam	100	35	56354.8	56354.8
41	Ammonia, FSA, AS mix	100	35	Ammonia: 1564.5 FSA: 7.4e-06 AS: 13328.0	14892.5
42	AS	100	35	13328.0	13328.0
43	Ammonia, FSA mix	100	35	Ammonia: 1564.5 FSA: 7.4e-06	1564.5

Documentation of Simulation

The Aspen simulation shown above was originally created based on our theoretical PFD that is shown in Figure 1. However, there were many key assumptions and adjustments that needed to be made for our Aspen simulation to be fully operational and generate ϵ -caprolactam. These adjustments resulted in the Aspen PFD not directly aligning with the theoretical PFD

shown in Figure 1. It should be noted that because constant changes needed to be made to the Aspen simulation in an attempt to generate product and optimize the process, it was decided that all calculations for this report (material balances, energy balances, and economic calculations) would be based on the theoretical PFD (Figure 1). As opposed to the one generated in Aspen which was subject to change at any time.

Key assumptions that were made that resulted in a fully operational Aspen simulation included assuming that the hydroxylamine was heated to liquid phase before entering R-2, which nullified the need for conveyor belt. Similarly, it was assumed that phenol was entering R-1 as a vapor, rather than a liquid, such that we had a vapor-vapor reaction, rather than a vapor-liquid reaction. It was also assumed that all hydroxylamine sulfate and initial by-products were separated in S1, stream 12. Lastly, it was assumed that there was 35 kg of palladium catalyst in R-1, which had a particle density of 1.2 g/cm³.

In terms of adjustments, the most prominent was that R-3 in the theoretical PFD needed to be changed into 3 reactors in series, rather than one single reactor, which are labeled as R-3-1, R-3-2, and R-3-3 in the Aspen simulation. In addition, an adjustment to our hydrogen feed was made from 2650 kg/hr in the theoretical PFD, to 2651 kg/hr in the Aspen simulation. Overall, because of making these assumptions and adjustments, our simulation is producing approximately 56,355 kg/hr which translates to about 544,000 tons/year. This means that the production rate obtained from the simulation is well over the target production rate of 500,000 tons/year, supporting the conclusion that this process is possible.

Errors and Learnings

The kinetics for the CSTRs that represent the third reactor (R-3-1, R-3-2, and R-3-3) were a prominent issue faced for this Aspen model. When the simulation was run, the results indicated that, due to kinetics not being completely accurate, there were negative flow rates of the limiting reactant. It was clear that obtaining a negative flow rate is not possible, as the minimum flow rate that is achievable is never less than zero. Therefore, to fix this, the kinetics from a CSTR that was operational were as the starting point. It is understood that the kinetics for all reactors in the process will realistically not be the same, however finding accurate kinetics values in literature posed a great challenge. The same kinetics values were therefore used for all CSTRs, which

allowed the simulation to run with no errors. Another error encountered during the formation of this Aspen file involved the bonds and bond structures for the less common compounds. Using their default setting would result in errors while running the simulation. To resolve this issue, the bond angles were calculated in the materials section in Aspen, and the chemical structures were redrawn as needed for proper calculations. Another error involved the distillation column (T-1). The distillation tower was initially set to “total condenser” which resulted in two liquid streams, one coming out of the top and one coming out of the bottom. It is known that a distillation column should result in a vapor leaving the top of the tower, and a liquid stream leaving the bottom of the tower, therefore our observations of having two liquid streams did not make practical sense. To resolve the issue, the tower was set to “partial condenser” which resulted in a vapor exiting the top of the tower and a liquid exiting the bottom. Lastly, there was an issue with the fuming sulfuric acid or oleum in reactor 3. Fuming sulfuric acid consists of both H_2SO_4 and SO_3 , which then reacts with water to form H_3O^+ , which is necessary for the Beckmann Rearrangement to form ϵ -caprolactam. When adding all these reactions to reactor 3, Aspen would crash. To correct this issue, we put these reactions in separate reactors in series, and the simulation worked and provided the necessary flowrates to meet our target production amount.

Material Balances

The following table displays the material balance information for our process. This includes temperature, pressure, the mass flow rate of our individual species, and the total flow rate. Other useful information that was required to calculate the following is located in Appendix A. The following abbreviations may be useful when viewing our table, as they were used for simplicity and organization of the table:

Cyclohexanone: ANON

Cyclohexanone oxime: CHOX

Hydroxylamine sulfate: HAS

Fuming sulfuric acid: FSA

Ammonium sulfate: AS

Table 3: Mass balance information corresponding to the theoretical PFD shown in Figure 1, including corresponding flowrates, temperatures, and pressures

Stream number	Chemical species	Temperature (C)	Pressure (bar)	Mass flow rate of chemical species (kg/s)	Total mass flow rate (kg/s)
1	Phenol	20	1	16.9	16.9
2	Hydrogen	20	1	0.7	0.7
3	Phenol	20	1	16.9	16.9
4	Hydrogen	20	1	0.7	0.7
5	ANON	450	2	17.6	17.6
6	ANON	450	2	17.6	17.6
7	ANON	85	2	17.6	17.6
8	ANON, CHOX, HAS mix	85	2	ANON: 1.87e-07 CHOX: 20.3 HAS: 8.4	31.9
9	ANON, CHOX, HAS mix	85	2	ANON: 1.87e-07 CHOX: 20.3 HAS: 8.4	31.9
10	HAS	25	1	14.4	14.4
11	HAS	25	1	14.4	14.4
12	ANON	160	1	13.6	13.6
13	ANON	450	1	13.6	13.6
14	ANON	450	1	13.6	13.6
15	CHOX, HAS mix	160	1	CHOX: 20.3 HAS: 4.1	26.1
16	CHOX, HAS mix	160	1	CHOX: 20.3 HAS: 4.1	26.1
17	CHOX, HAS mix	95	1	CHOX: 20.3 HAS: 4.1	26.1
18	CHOX	95	1	20.3	20.3

19	CHOX	95	1	20.3	20.3
20	Water	25	1	1.5	1.5
21	Water	25	1	1.5	1.5
22	HAS, water mix	95	1	HAS: 8.4 Water: 1.7	10.2
23	HAS, water mix	95	1	HAS: 8.4 Water: 1.7	10.2
24	HAS, water mix	100	1	HAS: 8.4 Water: 1.7	10.2
25	HAS, water mix	100	1	HAS: 8.4 Water: 1.7	10.2
26	Water	100	1.3	1.7	1.7
27	Water	25	1	1.7	1.7
28	Water	25	1	1.7	1.7
29	Water	25	1	1.7	1.7
30	Water	25	1	1.7	1.7
31	FSA	25	1	11.6	11.6
32	FSA	25	1	11.6	11.6
33	Caprolactam	90	1	15.7	15.7
34	Caprolactam	90	1	15.7	15.7
35	Caprolactam	100	1	15.7	15.7
36	Ammonia	-33	1	1.4	1.4
37	Ammonia	-33	1	1.4	1.4
38	Ammonia	100	1	1.4	1.4
39	AS	100	2	3.7	3.7
40	AS	100	2	3.7	3.7
41	AS	85	1	3.7	3.7
42	AS	60	1	3.7	3.7
43	AS	60	1	3.7	3.7
44	AS	25	1	3.7	3.7

45	Caprolactam	100	1	19.3	19.3
46	Caprolactam	100	1	19.3	19.3
47	Caprolactam	85	1	19.3	19.3

The mass flowrates presented in this report were determined using Aspen Plus. Aspen was utilized to model the full process based on our proposed reaction pathways, operating conditions, and separation steps. By inputting key parameters such as our process flow diagram, reaction conversions, component properties, and unit operation specifications, the software was able to perform mass and energy balances across the entire process. The temperature and pressure values were based on literature values. Our team recommends future research be conducted to verify that these properties support the efficient and reliable operation of the process. These values are critical for ensuring the technical feasibility and optimization of our design.

Energy Balances and Utility Requirements

This section will describe our heating, cooling, electrical power, and the other utilities required for this process. The following table describes the power and costs of our equipment requiring the greatest number of utilities, such as pumps, heat exchangers, and boilers.

Table 4: Pumps utility demand and cost breakdown

Equipment Number	Utility Usage (kW)	Annual utility cost (USD)
P-1	7.3	4,100
P-2	8.3	4,660
P-3	7.2	4,060
P-4	7.2	4,060
P-5	7.5	4,190
P-6	7.5	4,190
P-7	7.5	4,190
P-8	7.5	4,190
P-9	7.5	4,190
P-10	7.5	4,190

P-11	7.5	4,190
P-12	7.5	4,190
P-13	7.2	4,060
P-14	7.5	4,190
P-15	7.5	4,190
P-16	7.5	4,190
P-17	7.5	4,190

Table 5: Heat exchanger utility demand and cost breakdown

Equipment Number	Utility Usage (MJ/h)	Annual Utility Cost (USD)
HX-1	35500.0	112,000
HX-2	56.0	946
HX-3	290.0	910
HX-4	7.8	25
HX-5	56.0	946
HX-6	56.0	946
HX-7	56.0	946
HX-8	56.0	1,841
HX-9	56.0	946

Table 6: Evaporator utility demand and cost breakdown

Equipment Number	Utility Usage (MJ/h)	Annual Utility Cost (USD)
E-1	50.0	7.79

Most of our heating and cooling power comes from our heat exchangers and the evaporator. We have nine heat exchangers, and one evaporator. The utility used for our heat exchangers will be cooling water and low-pressure steam. To reduce our energy consumption and reduce capital costs, we have created a heat exchanger network using pinch analysis. The evaporator produces water vapor during the separation of our water and hydroxylamine sulfate.

Instead of discharging this vapor, our team proposes recovering the latent heat by condensing the steam in a heat exchanger. The recovered heat can be integrated into the heat exchanger network to preheat feed streams, reducing overall steam demand and lowering utility costs. The utility usage prior to creating a heat exchanger network comes to a total of 89,090.8 MJ/h, and the total annual utility cost for all equipment is \$3,330,000. After designing our network, we were able to reduce the utility usage to 36133.8 MJ/h, which is around a 60% decrease. The cost also decreased to \$2,440,000, making this option suitable for conserving energy and reducing costs.

The bulk of our electrical power comes from our pumps in the process. We have seventeen pumps total in our plant, and they all rely on electrical power to run. The total amount of power used is 127 kilowatts (kW). Based on a Capcost estimate, this gives us a total annual utility cost of \$71,220.

Equipment Cost Summary

The tables below contain the estimated purchase price of the equipment shown in our PFD. These estimated prices were obtained from CAPCOST software using a 2025 Chemical Engineering Plant Cost Index (CEPCI) of 542. The only equipment that is not estimated through CAPCOST are the gate control valves, which were estimated to have a purchased equipment cost of \$450 each (ECS [7]). Using CAPCOST's Bare Module Factor of 4.8, the Bare Module Cost was calculated to be \$2,160 for each gate control valve.

Table 7: Reactors cost summary

Type	Tag	Purchased Equipment Cost	Bare Module Cost	Base Equipment Cost	Base Bare Module Cost
Fermenter	R-1	\$115,000	\$460,000	\$115,000	\$460,000
Autoclave (x3)	(R-2 : R-4)	\$112,000	\$447,000	\$112,000	\$447,000

Table 8: Pumps cost summary

Type	Tag	Purchased Equipment Cost	Bare Module Cost	Base Equipment Cost	Base Bare Module Cost
Centrifugal	P-1	\$7,170	\$18,400	\$4,630	\$15,000
Centrifugal	P-2	\$7,460	\$19,200	\$4,810	\$15,600
Centrifugal (x2)	(P-3 : P-4)	\$7,150	\$18,400	\$4,610	\$14,900
Centrifugal (x5)	(P-5 : P-9)	\$7,220	\$18,500	\$4,660	\$15,100
Centrifugal	P-10	\$7,220	\$18,600	\$4,660	\$15,100
Centrifugal (x2)	(P-11 : P-12)	\$7,220	\$18,500	\$4,660	\$15,100
Centrifugal	P-13	\$7,150	\$18,400	\$4,610	\$14,900
Centrifugal (x4)	(P-14 : P-17)	\$7,220	\$18,500	\$4,660	\$15,100

Table 9: Other Equipment costs summary

Type	Tag	Purchased Equipment Cost	Bare Module Cost	Base Equipment Cost	Base Bare Module Cost
Empty Vertical Vessel (Distillation Tower)	T-1	\$393,000	\$1,350,000	\$283,000	\$1,150,000
Horizontal Storage Vessels (x8)	(V-1 : V-8)	\$191,000	\$574,000	\$191,000	\$574,000

Process Evaporator	E-1	\$2,260,000	\$4,820,000	\$2,260,000	\$4,820,000
Centrifugal Liquid- Liquid Separator	S-1	\$17,100	\$26,800	\$17,100	\$26,800
Floating Head Heat Exchangers (x9)	(HX-1 : HX-9)	\$170,000	\$385,000	\$62,400	\$205,000
Gate Control Valves (x48)	(GV-1 : GV-48)	\$450	\$2,160	\$450	\$2,160

Fixed Capital Investment Summary

Table 10: Total costs obtained from the equipment cost summary shown above

	Total Cost
Purchased Equipment Cost	\$6,301,820
Bare Module Cost	\$16,369,700
Bare Equipment Cost	\$5,201,490
Base Bare Module Cost	\$14,579,380
Module Cost	\$19,320,000
Grass Roots Cost	\$26,610,000
Lang Factor	4.74
Lang Factor Cost	29,900,000

For this Cash Flow Analysis, the **Fixed Capital Investment (FCI)** is the **Grass Roots Cost (\$26,610,000)**. This is the default FCI in CAPCOST's software.

These values were all obtained using CAPCOST's software. Equations used can be found from [55, p. 201, Figs. 7.15 and 7.16] and [55, p.184, Fig. 7.5]

Operating/Manufacturing Costs

Table 11: Total operating and manufacturing costs for this proposed process

Reagent Costs	<u>Annual Flow Rate:</u>	<u>Unit Cost:</u>
	Oleum: 55,609kg/h	[8] Oleum: \$0.13/kg
	Hydrogen: 60873 kg/h	[9] Hydroxylamine Sulfate:
	Water: 58932 kg/h	\$1.42/kg
	Phenol: 58,760 kg/h	[10] Hydrogen: \$1/kg
	Total Cost: \$1,268,844,948/Yr	[11] Water: \$0.0006/kg [12] Phenol: \$1.00/kg
Utility Costs	Total Cost: \$2,440,000/Yr	<u>Individual Utility Cost:</u>
		Electricity: \$79,009/Yr
		Low Pressure Steam:
		\$6,571/Yr
		Cooling Water: \$112,935/Yr
Labor Costs	53 operators needed	[13] Median Operator Salary:
	Total Cost: \$4,241,590/Yr	\$80,060/Yr
Taxation Rate	25%	20% Federal
		5% State
Waste Treatment	Total Cost:	
	\$1,664,400/Yr	

A more thorough explanation as to how these values were obtained can be found in Appendix B.

Overall Economic Analysis

The system design will have a two-year construction period, and a 13-year operating period. The sale price of ϵ -caprolactam will need to be \$3.00/kg to make a profit after a total of 10 years: 2 years for construction and 8 for operation. The discount rate was 13.74%. The equation used to

calculate IRR is: $0 = \sum_{n=0}^N \frac{C_n}{(1+IRR)^n}$, where N = holding period, n = number of periods, and IRR = Internal Rate of Return, which gives the discount rate when NPV is zero.

The equation used to calculate NPV is: $NPV = \sum_{t=0}^n \frac{R_t}{(1+i)^t}$, where NPV is net present value, R_t = Net cash inflow-outflows during a single period, t = number of time periods, and i = discount rate.

For our baseline case, we are using a palladium catalyst, and in our comparison case we are using a cheaper, though less effective nickel catalyst. To account for the cost of the catalyst, the cost of the mass needed of each catalyst for 15 years was added to the initial cost of land in Capcost. This shows a worst-case situation where all catalyst materials needed are purchased up front. The NPV is shown below for the two catalysts.

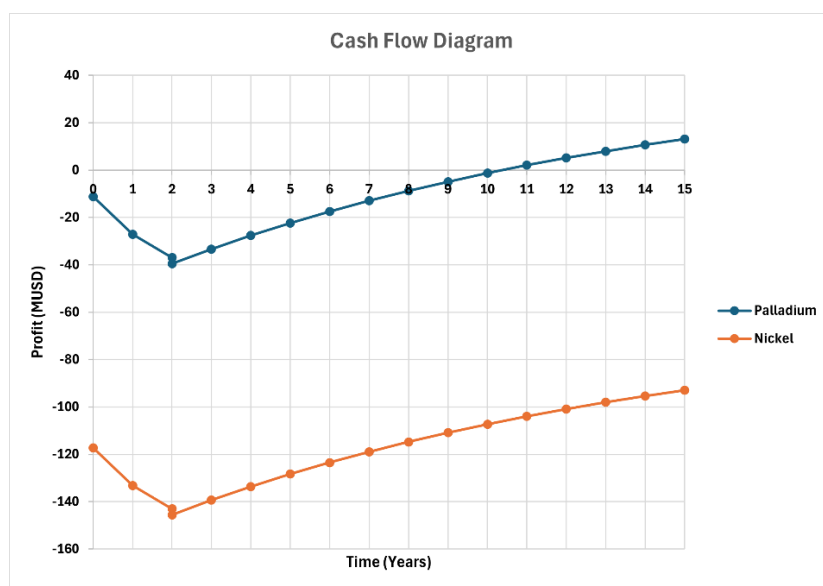


Figure 3: Comparison of palladium and nickel catalyst NPVs

The NPV of this system using a palladium catalyst is 18.4 million USD, where the NPV of the system with a nickel catalyst is -54.1 million USD, based on their mass costs [14-17]. This is due to the significantly higher amounts of material needed for nickel due to lower selectivity and catalytic performance. Thus, palladium is chosen as the catalyst material being used.

Conclusion and Recommendations

The proposed process for the production of ϵ -caprolactam from biomass-derived phenol is both technically and economically feasible. It is strongly recommended for implementation and development. The process achieves a capacity of approximately 544,000 US tons per year with a final product purity of 98.5%, meeting industry specifications while effectively reducing the need for petroleum-based feedstock.

Comprehensive economic analysis shows a positive net present value (NPV) of \$18.4 million over 15 years for the utilization of a palladium catalyst, which is far more desirable than the less selective nickel catalyst that yields a significantly negative NPV. The operation is designed with robust safety, sustainability, and efficiency in mind. For instance, a heat exchanger network that reduces energy consumption by 60% and reagent and water recycle systems to limit waste. Additionally, our site layout is designed to account for environmental risks such as high winds and power interruptions, which are common in New Mexico.

To continue the improvement of the process, we recommend future work on process optimization - especially in reactor design and catalyst efficiency, conducting a full life cycle assessment to more effectively quantify the environmental benefits, constructing a pilot plant for verification and scale-up, and ensuring early engagement with regulatory agencies and the local community. Overall, this project offers an appealing, renewable pathway to ϵ -caprolactam production and is both environmentally sustainable and economically viable.

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Appendices

Appendix A: Material Balances

Table 1: Density and viscosities of chemical species

Chemical species	Density (kg/m ³)	Viscosity (kg*m*s)
Phenol	1070 [18]	0.008 [19]
Hydroxylamine sulfate	1880 [20]	Not found
Hydrogen	0.08 [21]	8.988×10^{-8} [22]
Fuming sulfuric acid	1900 [23]	Not found
Water	1000	8.891×10^{-4} [24]

Cyclohexanone	947.8 [25]	0.00898 [26]
Cyclohexanone oxime	1130 [27]	0.192 [28]
Ammonia	0.73 [29]	1.862×10^{-4} [30]
Ammonium sulfate	1769 [31]	Not found
ϵ-caprolactam	1010 [31]	0.059 [31]

Appendix B: Operational Costs Calculation

The cost of our labor per year will be \$4,241,590. We calculated the amount of operators necessary with the equation:

$$N_{oL} = [6.29 + 31.7P^2 + 0.23(N_{np})]^{0.5} \text{ (Ch8)}$$

where N_{oL} is Number of Operators Per Shift, P is the number of processes handling transport of a particulate solid, N_{np} is the number of nonparticulate processing steps. Based on this calculation, we will need 11.68 operators per shift:

$$N_{oL} = [6.29 + 31.7(2)^2 + 0.23(15)]^{0.5} = 11.68.$$

The next equation calculates the total number of operators we need:

$$C_{oL} = 4.5(N_{oL}) \text{ (Ch8)}$$

where C_{oL} is the 4.5 operators needed per each required operator to cover all shifts in a year.

Using this equation, we will need 53 operators:

$$C_{oL} = 4.5(11.86) = 52.58$$

The 52.58 is rounded up to 53. The median salary for an operator is \$80,030 [39], and multiplied by the 53 operators we need, the \$4.2 million is our cost of labor. Our raw materials are phenol, hydrogen, water, sulfuric acid, and hydroxylamine sulfate, with prices per kg of \$1.00, \$1.70, \$0.50, \$1.00, and \$142.00 respectively [13-15]. The annual sum with all of our flow rates \$215,438,625,784, which was calculated using Capcost.